

Examiner
only

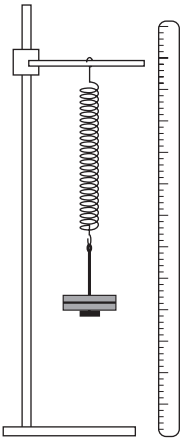
7. A Welsh company has been asked to design a toy that is able to launch a model plastic plane. They decide to use a spring mechanism in its design. Initially they carry out an experiment to investigate whether a spring is suitable for use in the toy.

Slotted masses are added to exert a force on the spring.

The resulting length of the spring is measured.

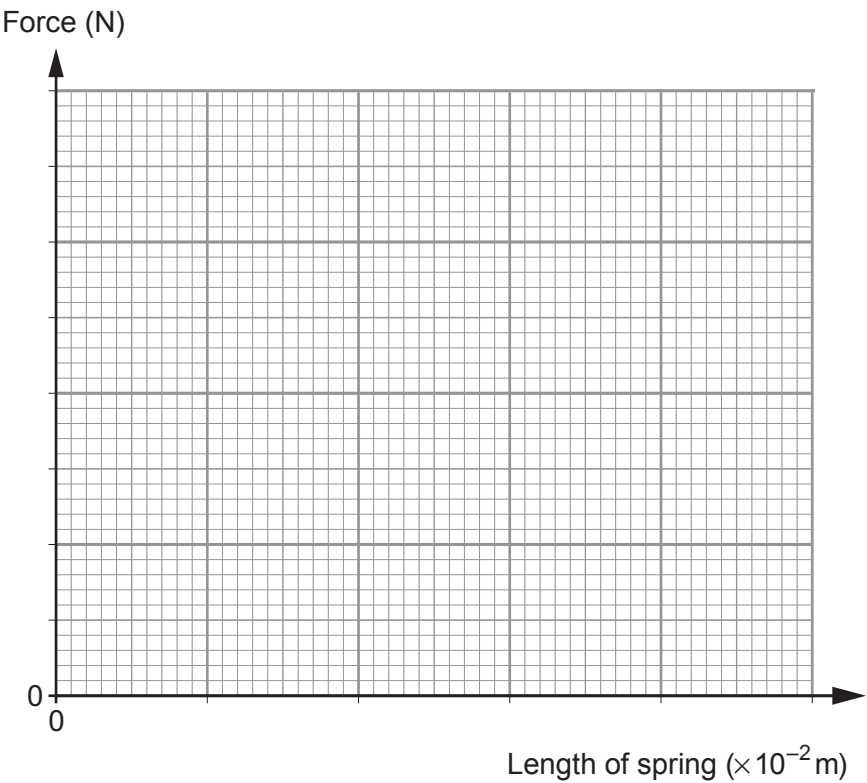
The results are shown in the table below.

Mass (g)	Force (N)	Length of spring ($\times 10^{-2}$ m)
100	1.0	7.5
200	2.0	10.0
300	3.0	12.5
600	6.0	20.0
700	7.0	22.5



- (a) (i) Plot the data on the grid below and draw a suitable line.

[4]



- (ii) State the length of the spring when no force is applied.

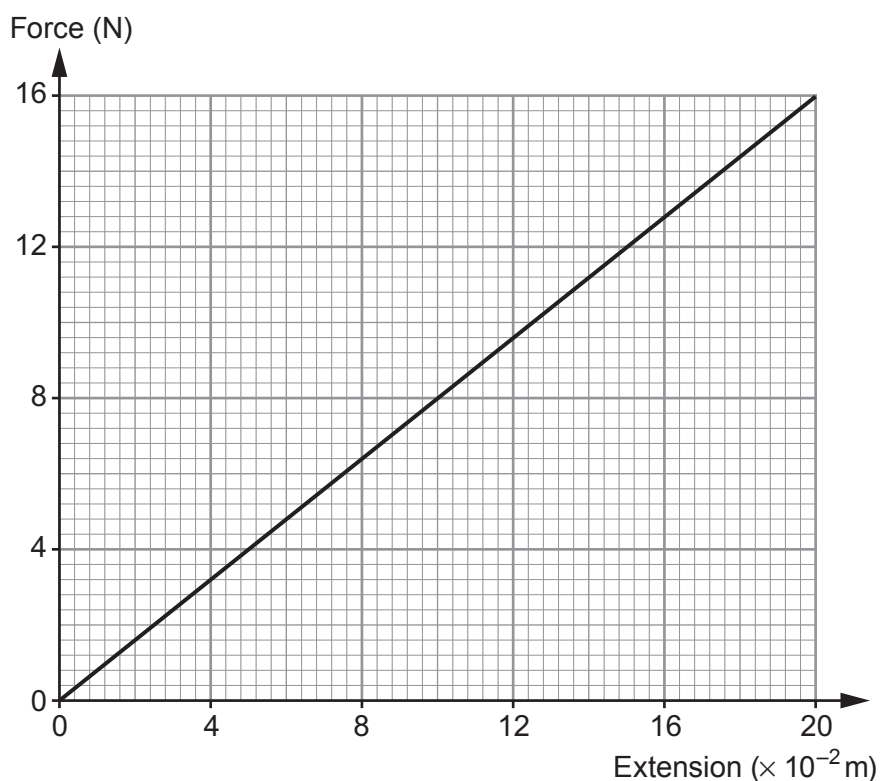
[1]

Length of spring = $\times 10^{-2}$ m



- (b) The design company decides to connect two of these springs in parallel with each other in the launch mechanism of the model plastic plane.

The graph shows how the extension of this spring system changes with force.



The two spring system is stretched and a model plastic plane of mass 55×10^{-3} kg is attached. A trigger is pressed, which releases one end of the spring system, and the plane is launched.

- (i) The two spring system is stretched to a maximum extension of 15×10^{-2} m. Use the graph and an equation from page 2 to calculate the work done stretching this spring system. Give an appropriate unit with your answer. [4]

Work done =

Unit



- (ii) Health and safety guidelines limit the launch velocity of the plastic plane to a maximum of 4 m/s. Using an equation from page 2 and your answer from (b)(i), calculate the launch velocity of this plane (mass = 55×10^{-3} kg) **and state** whether the launch system design meets this safety guideline. [3]

Launch velocity = m/s

- (iii) Explain why the actual launch velocity is always less than your calculated launch velocity. [2]

- (iv) A student suggests that a maximum momentum value would be more appropriate than stating a maximum velocity restriction for the launch. Explain why this would be an improvement to the health and safety guidelines. [2]

END OF PAPER

